Abb Muhaimeen

bsse1423@iit.du.ac.bd | +8801815890998 | LinkedIn: muhaimeen-alam | GitHub: muhaimeen90 | Portfolio-Website

EDUCATION

University of Dhaka – Institute of Information Technology — **Dhaka, Bangladesh**
Bachelor of Science in Software Engineering – CGPA: 3.81/4.00 — *Expected Graduation: September 2026*

Rajuk Uttara Model College — **Dhaka, Bangladesh**
Higher Secondary Certificate Examination – GPA: 5.00/5.00 — *December 2021*

Rajuk Uttara Model College — **Dhaka, Bangladesh**
Secondary School Certificate Examination – GPA: 5.00/5.00 — *April 2019*

RESEARCH EXPERIENCE

UIUC++ Summer Research Program, University of Illinois Urbana-Champaign
Remote Research Intern | Supervisors: Prof. Wing Lam, Prof. August Shi — *June 2026 – Present*

- Developing an LLM-based agentic tool that automatically implements Gherkin (Given/When/Then) step definitions and repairs Page Object Models, eliminating the manual coding step that Behavior-Driven Development workflows currently require of developers
- Designed an agent loop with tool support that explores the existing codebase (POMs, fixtures, XPaths), synthesizes the step-definition implementation, executes the resulting tests, and iterates on failures
- Integrated automated screenshot capture of executed scenarios so non-technical stakeholders can validate that a requirement was implemented correctly without a developer round-trip

University of Alabama
Remote Research Assistant | Supervisor: Dr. Rayhanur Rahman — *May 2026 – Present*

- Investigating how small language models perform on malware classification using the EMBER dataset
- Running controlled experiments across model scales and prompting strategies to quantify accuracy-versus-cost trade-offs against conventional ML baselines

2025-Multimodal Parameter-Efficient Fine-Tuning Group
Remote Research Intern — *September 2025 – November 2025*

- Collaborated remotely with an international research group on a project focusing on PEFT techniques
- Executed PEFT methods like LoRa, QLoRa etc and tested its adaptability for multi-modality

IIT HCI Community
Research Intern — *September 2024 – January 2025*

- Contributed to ongoing research projects in Human-Computer Interaction (HCI) domain
- Assisted in conducting a literature review and Focus Group Discussions for an HCI-related paper

TEACHING EXPERIENCE

Bangladesh AI Olympiad (BDAIO)
Mentor, Regional Stage — *2026*

- Mentored participants through the regional stage, guiding them on problem-solving approaches and core AI/ML concepts

Independent Private Tutor — *2022 – Present*

- Tutoring high school and college students across both Bangladesh National Curriculum and International (Edexcel) curriculum
- Prepared individualized lesson plans and adapted instruction to differing syllabus structures and assessment formats

PROFESSIONAL EXPERIENCE

Streams Tech Ltd
Research and Development Intern — *October 2025 – Present*

- Developed and deployed GenAI-powered modules for a production application, improving feature automation efficiency by 35% and reducing manual processing overhead

- Built practical Computer Vision pipelines (object detection, image analysis), increasing model accuracy by 12% and reducing inference latency by 25% through optimization and preprocessing
- Developed core features for a client-facing live election tracking platform deployed during the 2026 Bangladesh national election, delivering real-time results with [99]% uptime under peak traffic

PROJECTS

Malware Copilot *Supervisor: Dr. Rayhanur Rahman*

- Built an AI-assisted malware analysis workbench that explains what a sample executable does rather than only classifying it, targeting the black-box behavior and false-positive burden of existing ML detectors
- Coupled Ghidra decompilation output with LLM reasoning to summarize and trace pseudocode behavior for analysts
- Implemented a RAG pipeline over a MITRE ATT&CK vector database that maps observed behavior to technique IDs with cited justifications, rendered on an ATT&CK Navigator-style heatmap
- Added function-level code embedding similarity search against known malware corpora to flag risky samples and surface the closest matching malware family

CharityGuard

- Engineered a decentralized Web3 charity platform using Node.js and Next.js, integrating blockchain to secure donation workflows
- Deployed on the Polygon Amoy Testnet using Hardhat, delivering an end-to-end working prototype

Connect Four

- Developed a Connect Four game in Lua, integrating the Minimax algorithm for intelligent move selection
- Enhanced AI performance by 25% through fine-tuning the evaluation heuristics function for faster decision-making

CubeMate

- Built a fully functional Rubik's Cube Solver in C++ during my freshman year, without relying on external libraries
- Implemented the complete algorithm from scratch, achieving 10% better performance than library functions

ACTIVITIES & ACHIEVEMENTS

Bangladesh AI Olympiad (BDAIO)
Problem Setter *2026*

- Authored competitive problems for the national AI olympiad; one problem was selected for the regional round

IIT Software Engineers' Community
Office Secretary *September 2024 – September 2025*

- Serve on the executive body of a 50+ member organization of Software Engineering students, organizing industry-relevant seminars and nationwide technology events

Prothom Alo "Quarantine Short Film Fest 2020"
Winner *June 2020 – July 2020*

- Co-wrote the screenplay for a short film selected as a winning entry in a nationwide competition organized by Prothom Alo

Competitive Programming

- Codeforces Pupil (Max Rating: 1324)

SKILLS

Programming: Python, Java, C/C++, SQL, Node.js, React.js, Lua, JavaScript/Typescript, HTML/CSS, Solidity
Tools: Git/Github, Linux/UNIX, MySQL, MongoDB, Figma, Jupyter Notebook, Google Colab, IntelliJ